# Output Form (OF)

## Score: 1/5 — Serious concern | Confidence: high

***Benchmark:** LAILA* | ***Context:** Arab High-School Arabic Essay Feedback (Qatar Primary)*

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**Output Form definition:** Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

## Justification

The output-form mismatch is the most direct and explicitly documented gap. LAILA evaluates models using QWK between human and model numeric scores [Q58] with hyperparameter optimization on average QWK [Q59, Q60, Q61]; LLMs are constrained to JSON-formatted scores via the outlines library [Q190]. The deployment requires open-ended natural-language revision suggestions in MSA accessible to G10–12 students [elicitation A3, regional YAML output_function]. QWK on numeric scores cannot capture pedagogical usefulness, actionability, or comprehensibility of generated feedback. The authors themselves recognize this limit by prohibiting high-stakes use [Q104] and recommending 'fairness auditing and stakeholder consultation before real-world application' [Q106]. Web research confirms no Arabic rubric-to-natural-language feedback evaluation framework exists [WEB-12, WEB-13, WEB-23]. OF was marked HIGH priority. Even basic infrastructure for evaluating feedback quality (rubrics for actionability, register-appropriateness, MSA fluency) is absent from the benchmark.

| ID | Evidence |
|---|---|
| Q58 | “We evaluate model performance using QWK to measure agreement between human and model scores.” (p.8) |
| Q59 | “Hyperparameters are tuned via Bayesian optimization with the Tree-structured Parzen Estimator algorithm (Bergstra et al., 2011), implemented using Optuna's TPESampler.” (p.8) |
| Q60 | “We ran 20 trials with 5 startup trials and a fixed random seed of 11.” (p.8) |
| Q61 | “The best configuration, selected based on average QWK on the development set, was used for final evaluation on the test data.” (p.8) |
| Q190 | “For all the LLM experiments, we used vLLM for inference with the outlines library to enforce the JSON output format.” (p.18) |
| Q104 | “We explicitly prohibit its use for high-stakes educational decisions affecting individual students.” (p.10) |
| Q106 | “We recommend that any system developed using LAILA undergo fairness auditing and stakeholder consultation before real-world application.” (p.10) |
| WEB-12 | AES literature explicitly identifies feedback generation as an overlooked area (https://link.springer.com/article/10.1007/s10462-024-11017-5) |
| WEB-13 | AAEE Saudi system provides corrected essays but not trait-grounded revision suggestions (https://www.researchgate.net/publication/343796441_Automated_students_arabic_essay_scoring_using_trained_neural_network_by_e-jaya_optimization_to_support_personalized_system_of_instruction) |
| WEB-23 | No Arabic system bridges trait scores to actionable natural-language feedback for secondary students (https://www.sciencedirect.com/science/article/abs/pii/S0306457319304935) |

## Strengths

- QWK is the established gold-standard metric for ordinal scoring agreement [Q58]
- JSON output enforcement and three-seed averaging for LLM experiments improve numeric-output reproducibility [Q189, Q190]
- Authors transparently flag the limits of the current evaluation [Q104, Q106]

| ID | Evidence |
|---|---|
| Q58 | “We evaluate model performance using QWK to measure agreement between human and model scores.” (p.8) |

| | |
|---|---|
| **Q189** | “To account for variability in example selection, each experiment was repeated three times with random seeds 1, 11, and 42, and we report the average performance across runs.” (p.18) |
| **Q190** | “For all the LLM experiments, we used vLLM for inference with the outlines library to enforce the JSON output format.” (p.18) |
| **Q104** | “We explicitly prohibit its use for high-stakes educational decisions affecting individual students.” (p.10) |
| **Q106** | “We recommend that any system developed using LAILA undergo fairness auditing and stakeholder consultation before real-world application.” (p.10) |

## Checklist

### Checklist item definitions:

| ID | Assessment Question |
|---|---|
| **OF-1** | Does output modality match regional deployment needs? |
| **OF-2** | Assess text-to-speech availability for speech output |
| **OF-3** | Consider literacy rates and accessibility requirements |
| **OF-4** | Document form mismatches harming external validity |

### Checklist responses:

**OF-1**: No — benchmark output modality is structured numeric/JSON scores [Q58, Q181, Q190]; deployment requires open-ended natural-language MSA feedback. Direct mismatch.

**OF-2**: Not applicable — deployment is text-based; TTS not required.

**OF-3**: Literacy is high in target population (G10–12 students); MSA register-appropriateness for adolescents is the key accessibility consideration, on which the benchmark provides no signal.

**OF-4**: Documented form mismatch: QWK [Q58] cannot capture pedagogical usefulness, comprehensibility, or actionability of generated revision suggestions; benchmark explicitly prohibits operational deployment without independent validation [Q104, Q105].

| ID | Evidence |
|---|---|
| **Q58** | “We evaluate model performance using QWK to measure agreement between human and model scores.” (p.8) |
| **Q181** | “In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics.” (p.18) |
| **Q190** | “For all the LLM experiments, we used vLLM for inference with the outlines library to enforce the JSON output format.” (p.18) |
| **Q104** | “We explicitly prohibit its use for high-stakes educational decisions affecting individual students.” (p.10) |
| **Q105** | “Users must commit to: (1) preventing re-identification attempts, (2) avoiding demographic profiling or inference, and (3) refraining from deployment in operational assessment contexts without independent validation.” (p.10) |
| **WEB-12** | AES literature explicitly identifies feedback generation as an overlooked area (https://link.springer.com/article/10.1007/s10462-024-11017-5) |
| **WEB-13** | AAEE Saudi system provides corrected essays but not trait-grounded revision suggestions (https://www.researchgate.net/publication/343796441_Automated_students_arabic_essay_scoring_using_trained_neural_network_by_e-jaya_optimization_to_support_personalized_system_of_instruction) |

## Information Gaps

- Whether any supplementary Arabic feedback-generation evaluation rubric is in development
- Whether downstream LLM-generated feedback can be calibrated against LAILA scores

## Requires Expert Verification

- Pedagogical experts and Arabic-language teachers on what 'good' formative feedback looks like in MSA for G10–12

## Remediation

**Gap 1: Benchmark evaluates only numeric/JSON score agreement (QWK), providing no signal on natural-language feedback quality, actionability, or MSA register-appropriateness.**

**Recommendation:** Construct a supplementary feedback-generation evaluation: pair each LAILA essay with teacher-authored model revision feedback, then evaluate generated feedback on a multi-axis rubric (faithfulness to trait scores, actionability, MSA register, student comprehensibility). Pilot with Arab high-school teachers and G10–12 students before any deployment.